

## Comparing data sets



1 The number of goals scored by Team 1 and Team 2 in a football tournament are recorded in the following table:

- a Find the total number of goals scored by both teams in Match C.
- b Find the total number of goals scored by Team 1 across all the matches.
- c Calculate the mean number of goals scored by:
  - i Team 1
  - ii Team 2

| Match | Team 1 | Team 2 |
|-------|--------|--------|
| A     | 2      | 3      |
| B     | 4      | 2      |
| C     | 5      | 2      |
| D     | 3      | 5      |
| E     | 3      | 4      |

2 The table shows the scores of Student A and Student B in 5 separate tests:

- a Find the mean of the scores of:
  - i Student A
  - ii Student B
- b What is the mean of the combined scores of the two students?
- c What is the highest score overall and which student obtained that score?
- d What is the lowest score overall and which student obtained that score?

| Test | Student A | Student B |
|------|-----------|-----------|
| 1    | 79        | 77        |
| 2    | 74        | 83        |
| 3    | 86        | 72        |
| 4    | 89        | 94        |
| 5    | 98        | 97        |

3 The salaries of men and women working the same job at the same company are given below:

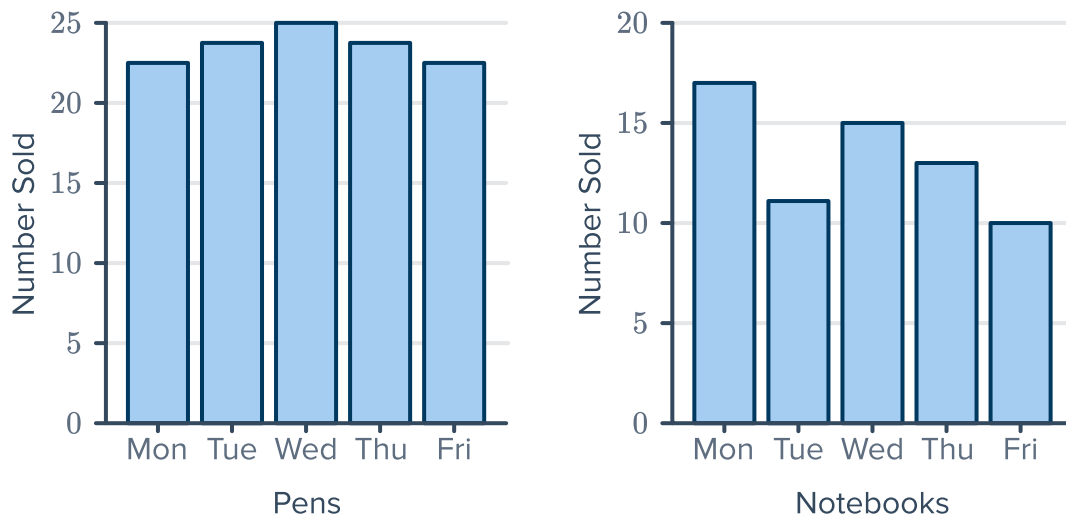
|       |        |        |        |        |        |        |        |
|-------|--------|--------|--------|--------|--------|--------|--------|
| Men   | 80 000 | 80 000 | 75 000 | 80 000 | 75 000 | 70 000 | 80 000 |
| Women | 70 000 | 70 000 | 75 000 | 70 000 | 70 000 | 80 000 | 75 000 |

- a Copy and complete the table, giving your answers correct to two decimal place where appropriate:

|        | Men | Women |
|--------|-----|-------|
| Mean   |     |       |
| Median |     |       |
| Mode   |     |       |
| Range  |     |       |

- b Which gender seems to be paid better? Justify your answer.

- 4 A stationery shop kept a weekly record of  number of pens and notebooks sold:



- Find the number of pens sold on Tuesday.
- Find the number of notebooks sold on Friday.
- Find the total number of notebooks sold during the week.
- Find the total number of pens sold during the week
- Find the percentage of pens sold on Thursday, rounding your answer to two decimal places.
- Calculate the mean number of notebooks sold per day. Round your answer to one decimal place.
- Calculate the mean number of pens sold per day. Round your answer to one decimal place.
- Which is the better selling product? Justify your answer.

- 5 The runs scored by each player in a cricket match are displayed below:

- Team A: 63, 0, 81, 22, 39, 66, 9, 18, 3, 3, 5
- Team B: 21, 26, 18, 69, 62, 26, 5, 28, 2, 1, 2

- Find the mean runs of the following teams, rounded to one decimal place:
  - Team A
  - Team B
- Which team scored more runs?

6 In a standardised test out of 100, the students in two classes got the results displayed below:

- Class A: 70, 64, 94, 78, 79, 61, 69, 93, 64, 79, 64, 71, 82, 68, 66
- Class B: 86, 68, 51, 80, 63, 87, 79, 89, 62, 79, 84, 77, 67, 80, 75

a Find the median mark for:

i Class A

ii Class B

b Using the medians, which class did better on the test?

7 The monthly rainfall for two cities across a year is displayed below:

- City A: 122, 120, 121, 67, 79, 24, 8, 46, 27, 67, 80, 128
- City B: 51, 66, 33, 78, 79, 92, 78, 99, 41, 60, 26, 56

a Find the range in monthly rainfall for:

i City A

ii City B

b Which city had more consistent rainfall throughout the year?

8 Marge grows two different types of bean plants. She records the number of beans that she picks from each plant for 10 days. Her records are shown below:

- Plant A: 10, 4, 4, 5, 7, 10, 3, 3, 9, 10
- Plant B: 8, 7, 5, 5, 9, 7, 8, 7, 5, 6

a What is the mean number of beans picked per day for Plant A? Round your answer to one decimal place.

b What is the mean number of beans picked per day for Plant B? Round your answer to one decimal place.

c What is the range for Plant A?

d What is the range for Plant B?

e Which plant produces more beans on average?

f Which plant has a more consistent yield of beans?

9 The weight (in kilograms) of a group of men and women were recorded and presented in a stem and leaf plot as shown:

| Women     |   | Men       |
|-----------|---|-----------|
| 7 6 3     | 5 |           |
| 8 6 3 1 1 | 6 | 2 3 3 8 9 |
| 3 1       | 7 | 1 2 4 8   |
|           | 8 | 2         |

a Find the mean weight of the group of men.

b Find the mean weight of the group of

- c Which group is heavier?



Key:  $8|3 = 83$

10 The stem and leaf plot shows the batting scores of two cricket teams, India and Sri Lanka:

- a Find the highest score in:

- i India                      ii Sri Lanka

- b Find the mean score of:

- i India                      ii Sri Lanka

- c Calculate the combined mean of the two teams.

| India     |   | Sri Lanka |
|-----------|---|-----------|
| 9 6       | 3 | 0 2 5 6 9 |
| 9 7 3 1 1 | 4 | 2 7       |
| 7 2 0     | 5 | 6 8       |
|           | 6 | 3         |

Key:  $8|3 = 83$

11 The stem and leaf plot shows the test scores of two classes, A and B:

- a Find the highest score in:

- i Class A                      ii Class B

- b Find the mean score of:

- i Class A                      ii Class B

- c Calculate the combined mean of the two classes, correct to two decimal places.

| Class A   |   | Class B |
|-----------|---|---------|
| 7 3 2     | 6 | 2 7     |
| 9 8 7 1 0 | 7 | 1 3 6 7 |
| 3 0       | 8 | 1 2 4 6 |
|           | 9 | 1 8     |

Key:  $8|3 = 83$

12 The stem and leaf plot shows the number of books read in a year by a random sample of university and high school students:

- a Interpret the lowest score for the University students.

- b Compare the medians of both groups of students.

- c For which student group(s) is the mean greater than the median?

| University Students |   | High School Students |
|---------------------|---|----------------------|
| 7                   | 0 |                      |
| 6 6 3               | 1 | 0 0 3 5              |
| 4 3 2 1             | 2 | 1 2 4 4 6            |
| 9 8 8 6             | 3 | 1 8 9                |
| 8 2                 | 4 | 0 1                  |
|                     | 5 |                      |
|                     | 6 |                      |
| 3                   | 7 |                      |

Key:  $4|1|2 = 14$  books and 12 books

13 The back to back stem and leaf plot shows the amount of cash (in dollars) carried by a random sample of teenage boys and girls:

- Which group carried more cash?
- Find the median amount of cash that the boys carried
- Find the median amount of cash that the girls carried.
- Which group's distribution is roughly bell shaped?
- Which group has more variation in the amounts of cash?
- Were there any outliers in the boys' amounts? If so, what are the value(s)?
- Were there any outliers in the girls' amounts? If so, what are the value(s)?

| Boys        |   | Girls         |
|-------------|---|---------------|
| 7           | 0 |               |
| 1           | 1 | 1             |
| 5 4 1       | 2 | 2 6 8         |
| 8 5 4       | 3 | 3 4 4 6 6 8 9 |
| 9 8 2 2 2 1 | 4 | 3 4 6         |
| 9 7 4 3     | 5 | 4             |
| 8 5 2       | 6 |               |
| 3 1         | 7 |               |

Key: 1|2|2 = \$21 and \$22

14 The following back to back stem and leaf plot shows the length (in minutes) of a random sample of phone calls made by Sharon and Tricia:

- Who made a 14 minute phone call?
- Who has the higher median?
- Is Sharon's mean greater than her median?
- Is Tricia's mean greater than her median?

| Sharon    |   | Tricia |
|-----------|---|--------|
| 3         | 1 | 3 4    |
| 7 6 4 3 2 | 2 | 6 7 8  |
| 9 8       | 3 | 2 4    |
| 4 3       | 4 | 1 2    |
| 7 6       | 5 | 6 7 8  |

Key: 2|2|6 = 22 and 26

15 The back to back stem and leaf plot shows the number of pieces of paper used over several days by Charlie's and Dylan's students:

- Did Charlie's students use 7 pieces of paper on any day?
- Who's class had the higher median?
- Is the median greater than the mean in both groups?

| Charlie's students |   | Dylan's students |
|--------------------|---|------------------|
| 7                  | 0 | 7                |
| 3 2 1              | 1 | 3                |
| 8                  | 2 | 8                |
| 4 3 2              | 3 | 3 4              |
| 9                  | 4 | 5 6 7            |
| 2                  | 5 | 2 3              |

Key: 1|1|3 = 11 and 13

16 The back to back stem and leaf plot shows the number of desserts ordered at Hotel A and Hotel B over several randomly chosen days:

- a Interpret the lowest score for Hotel A.
- b Which hotel's median is higher?
- c Is the mean greater than the median in both groups?

| Hotel A |   | Hotel B |
|---------|---|---------|
| 3       | 0 |         |
| 4 3 2   | 1 | 3 4     |
| 7 6     | 2 | 7       |
| 4 3     | 3 | 3 4     |
| 6       | 4 | 6 7     |
| 2       | 5 | 2 3 4   |

Key: 2|1|3 = 12 and 13

17 The median house price in Oklachusetts is \$750 000 with a mean price of \$800 000 and the median house price in Brockway is \$750 000 with a mean price of \$680 000.

Which town is most likely to have some very expensive houses? Explain your answer.

18 Two Chemistry classes, each with 15 students, sit a 10-question multiple choice test, each with four possible answers (only one of which is correct). Their class results, out of 10, are below:

- Class 1: 3, 2, 1, 1, 2, 5, 2, 2, 2, 3, 2, 1, 5, 5, 4
- Class 2: 10, 6, 9, 8, 10, 9, 7, 7, 6, 8, 8, 10, 8, 9, 8

- a Complete the following table, rounding your answers to one decimal place when necessary:

|         | Mean | Median | Mode | Range |
|---------|------|--------|------|-------|
| Class 1 |      |        |      |       |
| Class 2 |      |        |      |       |

- b Which class was more likely to have studied for their test? Explain your answer.

19 The hours of sleep per night for two people over a two week period are shown below:



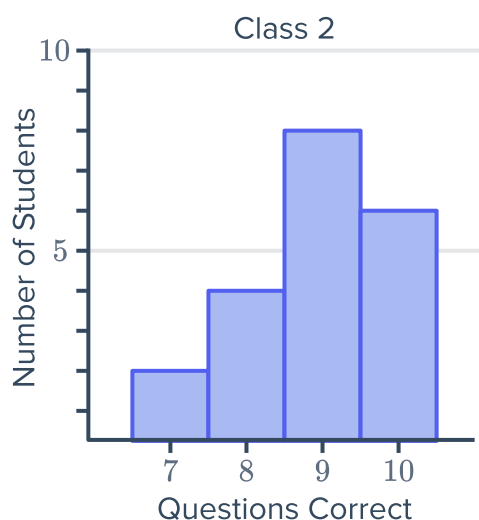
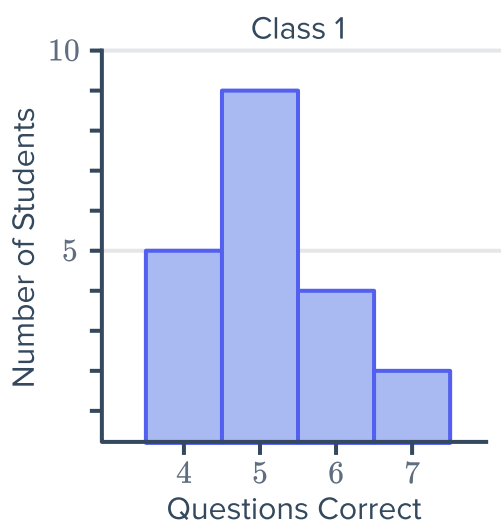
- Person A: 10, 9, 6, 6, 7, 8, 5, 6, 9, 9, 7, 5, 10, 9
- Person B: 7.5, 7.5, 8, 7.5, 7, 8, 7.5, 7, 7, 7, 7, 7, 8, 8

a Complete the following table, rounding your answers to one decimal place when necessary:

|          | Mean | Median | Mode | Range |
|----------|------|--------|------|-------|
| Person A |      |        |      |       |
| Person B |      |        |      |       |

- b Which person seems to be the least consistent in their sleep habits?
- c Which statistical piece of evidence supports your answer in part (b)?
- d Which person had the most sleep over the 14 nights?
- e Which statistical piece of evidence supports your answer in part (d)?

20 Two Science classes, each with 20 students, were given a 10 question True/False test. The results for each class are shown below:

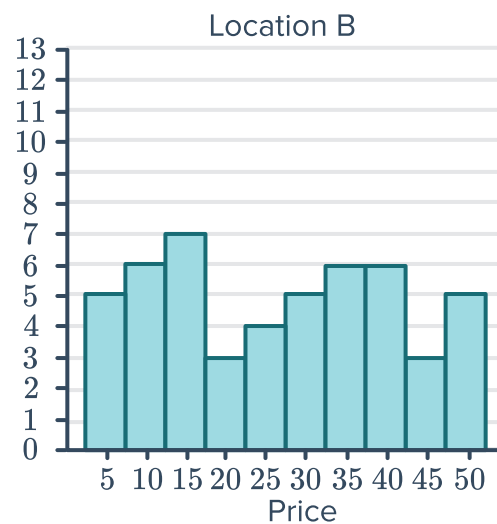
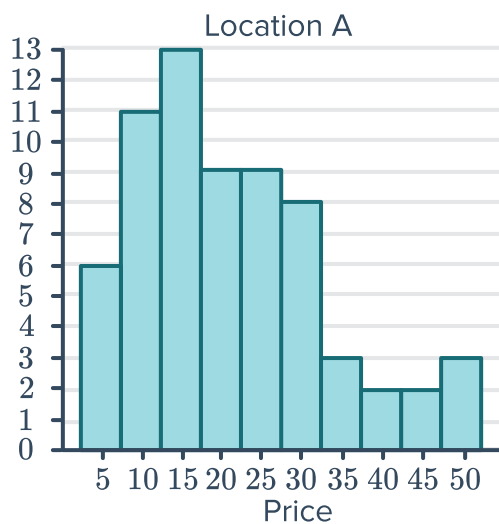


- a Calculate the mean for Class 1 correct to one decimal place.
- b Calculate the mean for Class 2 correct to one decimal place.
- c Find the range for Class 1.
- d Find the range for Class 2.
- e Do you think Class 1 studied for their test? Justify your answer.
- f Do you think Class 2 studied for their test? Justify your answer.



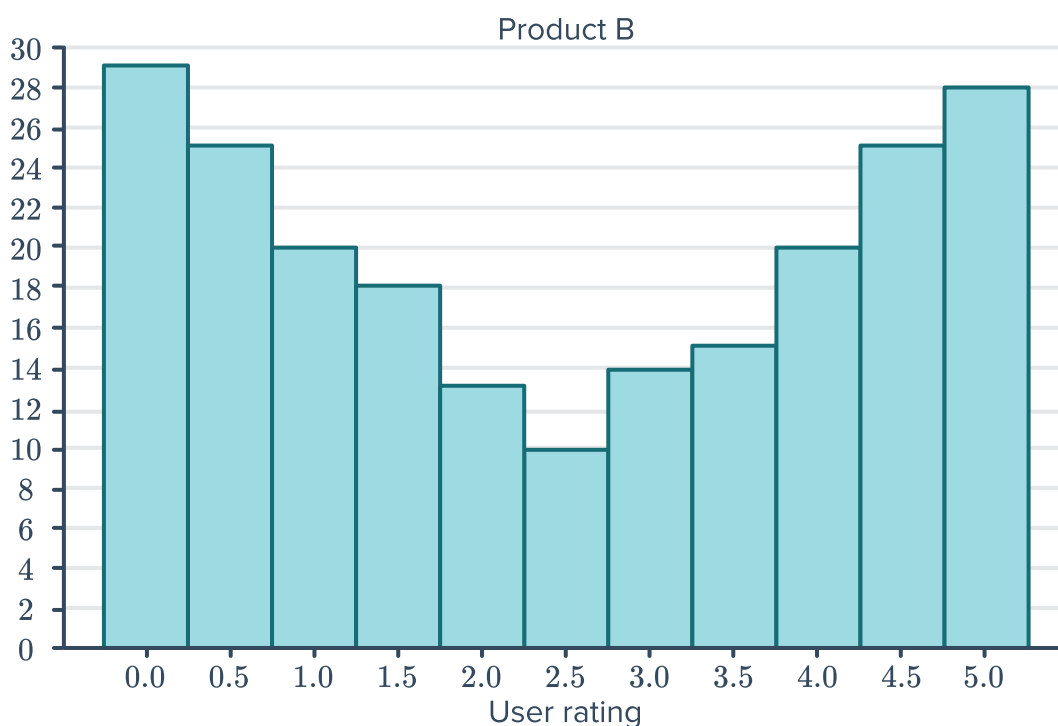
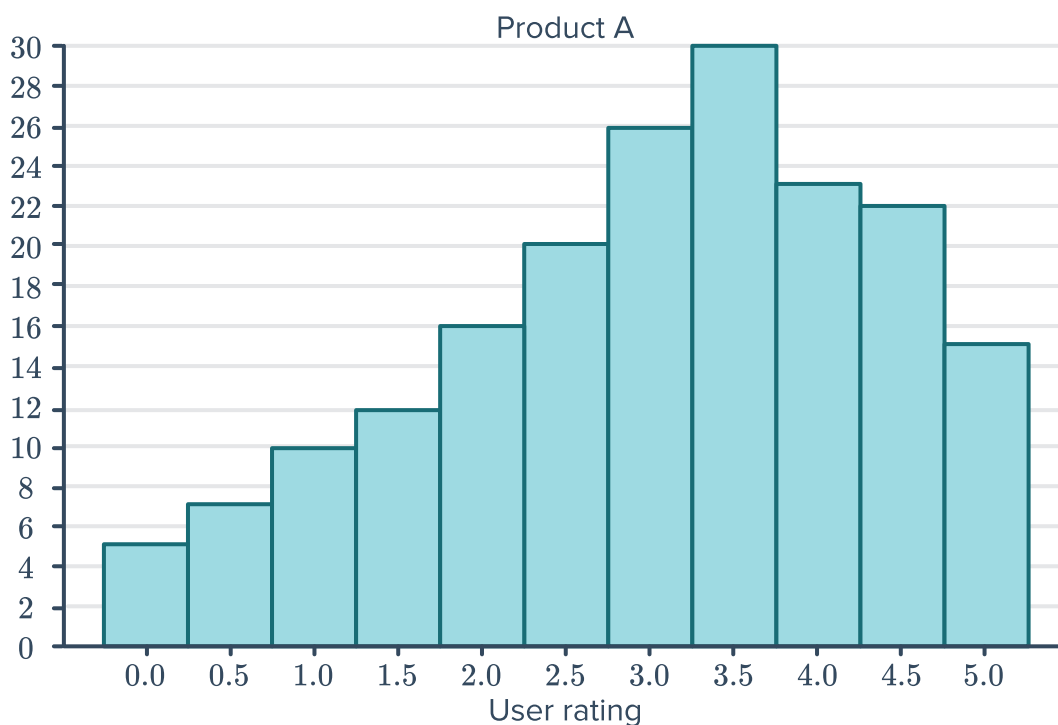


23 The prices of sales at two locations of a shop are recorded in the histograms below:



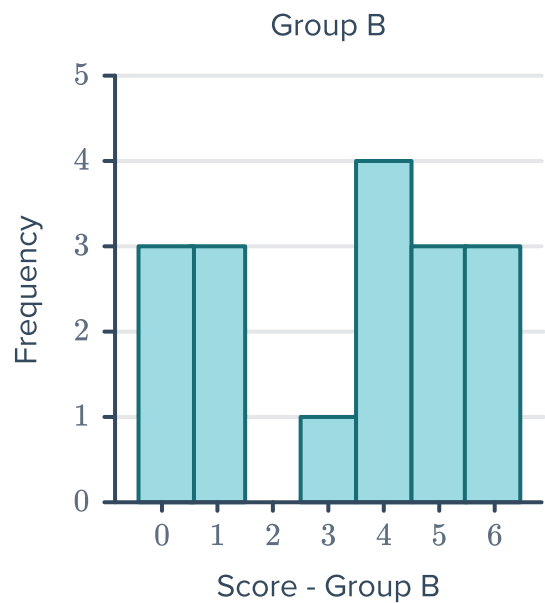
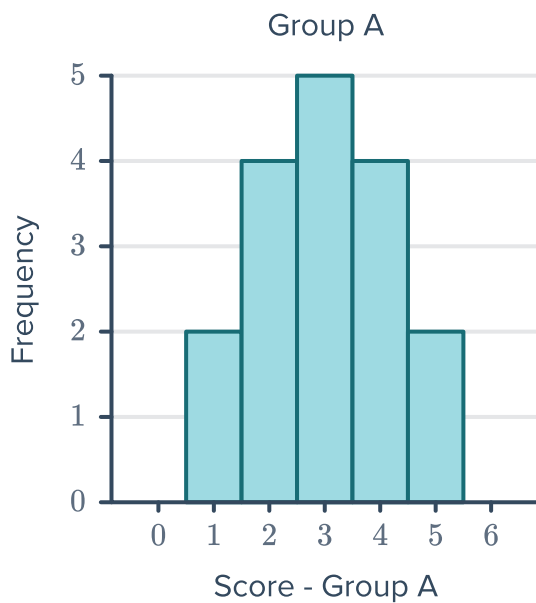
- a Find the mean sale price for:
- i Location A
  - ii Location B
- b Which location made more sales?
- c Which location had a higher average sale price?

24 An online shopping website records the user ratings for two similar products in the histograms below:



- a Find the median user rating for:
- i Product A
  - ii Product B
- b According to the website's rating system, a perfect rating is 5.0. Which product got more perfect ratings?
- c According to the website's rating system, a positive rating is greater than 3. Which product got more positive ratings?

25 The following histograms show the season results of two soccer groups, Group A and Group B, and the number of games (frequency) in which they scored a certain number of goals:



- a Find the mode for Group A.
- b Find the mode for Group B.
- c Find the range for Group A.
- d Find the range for Group B.
- e Which group scored the lowest total number of goals during the season?
- f Which group has the most varied results?